

The Assumptions of OLS: Multiple Regression

Hypothesis Testing: Multivariate Regression

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Where We Left Off

Models, technical interpretations, and substantive interpretations

- Models are systems of relationships presented in math-y terms.
- Technical interpretations: technical, math-y details
- Substantive interpretations: overall conclusions in context.

Regression gives us precise relationships and helps us predict outcomes

- Linear (OLS) regression fits a line that minimizes all of the “errors”
- $Y_i = \alpha + \beta_i X_i + \epsilon_i$ is just like $y = mx + b$ (plus error).
- Linear regression requires a continuous/interval DV.
- H_0 : the slope of the predictor (IV) is not different from 0 ($\beta_1 = 0$)
- H_1 : the slope of the predictor (IV) is different from 0 ($\beta_1 \neq 0$)

R-Square gives us an idea of the model's “goodness of fit”

- It behaves like a correlation (we interpret it similarly)
- It tells us how much of the variation in the DV the model explains.

Multivariate Regression

When we want to include multiple IVs in our regression model, we're doing a *multivariate* analysis.

Remember our equation?

$$Y = \alpha + \beta_i X_i + \epsilon_i$$

If we have multiple independent variables, we have multiple X's.

Let's say we have 3 IVs: X_1 , X_2 , and X_3

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon_i$$

How to Pick Multiple IVs

An overwhelming majority of regression analyses are multivariate.

But how do scholars choose which variables to analyze?

- Theoretical reasons
 - Multiple hypotheses to test
- Logical reasons
 - Control variables to account for confounding factors
- Mathematical reasons
 - Assumptions of OLS

A Partial Effect: when we have multiple IVs in the same model, each individual relationship constitutes a partial effect.

- Each partial effect contributes to the whole model, so including different variables can sometimes influence other variables (e.g., changes their effect, significance, etc.).

Control variables

Control variables are variables that we're not specifically interested in, but we suspect also relate to the DV in the background.

Why do we include control variables?

- To help isolate the main IV-DV relationship
- To prevent wrongly attributing results to the main IV(s)
- To reduce “noise” in the model from omitted variables

For example: if we want to look at how income influences ideology, we might control for race or region, too, to make sure that they're accounted for as important background factors that influence ideology.

How many IVs Is Too Many IVs?

Degrees of freedom:

- For any regression to work properly, we need to have fewer IVs than we have observations.

Avoid “kitchen sink” models:

- If everything is included in the model, it becomes more difficult to interpret and less clear that our variables matter (maybe they're just picking up random noise).
- We have to have a hypothesis (and null hypothesis) for every substantive variable we include...
- Sometimes including additional variables can *decrease* the model's explanatory power

Assumptions of OLS

OLS regression relies on a few key assumptions to make sure we can accurately and confidently interpret our results.

1. Linearity
2. Representative sampling
3. No multicollinearity (or perfect collinearity)
4. No heteroskedasticity
5. Independence (no autocorrelation)
6. Normally distributed errors, around 0.

The math will (usually) still work if we violate these assumptions, but our results will be called into question.

Multicollinearity

Double-counting related variables can cause problems with large variances and inverted signs of the coefficients (so interpretation is harder).

Multicollinearity is a situation where one of our independent variables can be predicted from the other variables, too!

We also have to worry about “**perfect collinearity**”, which means including multiple variables that completely determine one another.

- If we include a variable for POLS major, we can't also include a *separate* variable for Non-POLS major (because it perfectly predicts POLS major and prevents the math from running).

Solution: check correlations among the IVs, and exclude some “base category” from the model when we create multiple “dummy” variables.

Heteroskedasticity

If the variance among errors is not consistent, it will make the standard errors and confidence intervals too narrow or too wide, which undermines trust in our results.

Heteroskedasticity is a big scary word that describes an inconsistent variance is among the errors.

Solution: redefine the variable, weight the variable, transform the variable to make homoskedastic errors.

Independence and autocorrelation

OLS assumes that observations and variables are independent of one another, which means that they shouldn't cause each other.

Autocorrelation means correlation with itself.

- This happens a lot in data over time, when the outcome last year explains the outcome this year, for example.
- This also happens when we measure the same thing in multiple ways and include them all in the model.

Solution: really think through your theory and don't include multiple measures of the same thing.

Normally distributed errors

Normally Distributed Errors refers to the assumption that mean of all the error terms is zero, and that they're more or less normally distributed

If they're not, then we can't trust our predictions.

Solution: Cry. Just take my word for it. (It helps).

Remember: Transforming Variables

If an independent variable is really skewed to begin with, sometimes it's a good idea to transform it for a regression.

Here are some common transformations you'll see in academic publications:

- Natural log $\ln(x)$: positive (right) skew
- Roots/reciprocals $\sqrt{x}$ or $1/x$: positive (right) skew
- Square/cubes x^2 or x^3 : negative (left) skew
- Standardization $(x - \text{mean})/\text{standard deviation}$: removing different units for comparison

Do non-normally distributed IVs need to be transformed for OLS?

- Nope.
- We're talking about errors rather than variables, so this is not required, despite belief among some scholars.
- Sometimes we make the transformations, though, to make interpretations stronger, clarify patterns, or *attempt* to normalize the residuals, though.